

1 Introduction

Today we will be covering basic definitions, equalities, and inequalities of information theory. Perhaps it is the most important lecture.

2 Review

X is a discrete random variable (finite number of states) and entropy is defined as

$$H(X) = - \sum_a P(X = a) \log P(X = a) = -\mathbb{E}_X \log P(X)$$

Shannon's entropy is the only function that satisfies the grouping axiom and 3 additional, very reasonable, axioms.

If X is a random variable over n elements, then

$$0 \leq H(X) \leq \log n$$

2.1 Joint Entropy

For random variables X, Y then

$$H(X, Y) = -\mathbb{E}_{X, Y} \log P(X, Y)$$

and it generalizes over any number of variables

$$H(X_1, \dots, X_n) = -\mathbb{E}_{X_1, \dots, X_n} \log P(X_1, \dots, X_n)$$

2.2 Conditional Entropy

For random variables X, Y , then

$$H(X|Y) = \sum_b P(Y = b) \cdot H(X|Y = b) = -\mathbb{E}_{X, Y} \log P(X|Y)$$

Intuitively, it is the uncertainty remains in X that remains after seeing Y .

3 Fundamental Equalities and Inequalities

What can we say about the relations between:

$$H(X), H(Y), H(X|Y), H(Y|X), H(X, Y)$$

3.1 Fundamental Inequality of Conditional Entropy

For random variables X, Y

$$0 \leq H(X|Y) \leq H(X)$$

The first inequality is trivial, and the second is proved by Jensen inequality.

Intuitively, the uncertainty remaining in X after seeing Y is smaller than the original uncertainty in X . The uncertainty can only decrease by revealing Y . Also the uncertainty is always larger than 0.

$H(X|Y) = H(X)$ if and only if X, Y are independent. It's not hard but we will not prove.

3.1.1 Notable Example

Note that for specific b , it could be the case the $H(X|Y = b)$ could be larger than $H(X)$. The following table (1) is a good example.

	$X = 0$	$X = 1$
$Y = 0$	1/4	1/4
$Y = 1$	1/2	0

Table 1: Joint Probability Distribution Table

We have the following values

1. $H(X|Y) = \frac{1}{2}$
2. $H(X|Y = 0) = 1$
3. $H(X) = H(\frac{3}{4}, \frac{1}{4}) < 1$

3.1.2 Proof of the Second Inequality

The idea is to rework $H(X|Y)$ to get it into a state where we can apply Jensen's inequality, and then work back to $H(X)$

$$\begin{aligned}
H(X|Y) &= - \sum_{a,b} P(X=a, Y=b) \log P(X=a|Y=b) \\
&= - \sum_{a,b} P(X=a, Y=b) \log \frac{P(X=a, Y=b)}{P(Y=b)} \quad \text{via the probability chain rule} \\
&= \sum_{a,b} P(X=a, Y=b) \log \frac{P(Y=b)}{P(X=a, Y=b)} \\
&= \sum_{a,b} P(X=a) \frac{P(X=a, Y=b)}{P(X=a)} \log \frac{P(Y=b)}{P(X=a, Y=b)} \\
&= \sum_a P(X=a) \sum_b \frac{P(X=a, Y=b)}{P(X=a)} \log \frac{P(Y=b)}{P(X=a, Y=b)} \\
&\leq \sum_a P(X=a) \log \left(\sum_b \frac{P(X=a, Y=b)}{P(X=a)} \cdot \frac{P(Y=b)}{P(X=a, Y=b)} \right) \text{JENSEN} \\
&= \sum_a P(X=a) \log \left(\sum_b \frac{P(Y=b)}{P(X=a)} \right) \\
&= \sum_a P(X=a) \log \left(\frac{1}{P(X=a)} \right) \\
&= - \sum_a P(X=a) \log P(X=a) = H(X)
\end{aligned}$$

3.2 Chain Rule

The most beautiful formula

$$H(X, Y) = H(Y) + H(X|Y)$$

It's very intuitive! The uncertainty in (X, Y) is the uncertainty in Y plus the uncertainty remaining in X after revealing Y .

It's a real treasure and is used many times in applications.

We could have defined $H(X|Y)$ by this formula

$$H(X|Y) = H(X, Y) - H(Y)$$

The chain rule is the most important property of the entropy function!

3.2.1 Do we need to prove the chain rule?

The chain rule is just the grouping axiom!

We can visualize the joint distribution of X, Y as

	X			
Y	$p_{1,1}$	$\cdots$	$p_{1,n}$	q_1
	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	$p_{n,1}$	$\cdots$	$p_{n,n}$	q_n

By grouping the rows,, we have

$$H(p_{1,1}, \dots, p_{n,n}) = H(q_1, \dots, q_n) + \sum_{i=1}^n q_i \cdot H\left(\frac{p_{i,1}}{q_i}, \dots, \frac{p_{i,n}}{q_i}\right)$$

$$H(X, Y) = H(Y) + H(X|Y)$$

But we never proved the grouping axiom, so we do need to prove it.

3.2.2 Proof of the Chain Rule

$$P(X, Y) = P(Y) \cdot P(X|Y)$$

$$\log P(X, Y) = \log P(Y) + \log P(X|Y)$$

$$\mathbb{E}[\log P(X, Y)] = \mathbb{E}[\log P(Y)] + \mathbb{E}[\log P(X|Y)]$$

$$H(X, Y) = H(Y) + H(X|Y)$$

3.2.3 Chain Rule for $X_1, \dots, X_n$

For random variables $X_1, \dots, X_n$ we have

$$H(X_1, \dots, X_n) =$$

$$H(X_1) + H(X_2|X_1) + H(X_3|X_1, X_2) + \cdots + H(X_n|X_1, \dots, X_{n-1})$$

3.3 Subadditivity

It's just

$$H(X), H(Y) \leq H(X, Y) \leq H(X) + H(Y)$$

Equality in the second inequality if and only if X, Y are independent.

It's very intuitive! The uncertainty in (X, Y) is more than the uncertainty in X or Y , separately, but no more than the sum of uncertainties as we can reveal X and simultaneously reveal Y

3.3.1 Proof

Follows by:

$$H(X, Y) = H(Y) + H(X|Y) \quad \text{and} \quad 0 \leq H(X|Y) \leq H(X)$$

$$H(X, Y) = H(X) + H(Y|X) \quad \text{and} \quad 0 \leq H(Y|X) \leq H(Y)$$

The first inequality is quite straight forward since $0 \leq H(X|Y), H(Y|X)$. The second one follows from plugging $H(X|Y) \leq H(X)$ into $H(X, Y) = H(Y) + H(X|Y)$.

3.3.2 Subadditivity for $X_1, \dots, X_n$

$$H(X_1, \dots, X_n) \leq \sum_i H(X_i)$$

3.4 Equalities and Inequalities for Conditional Entropy

All equalities and inequalities that we have learned (and more generally all equalities and inequalities that involve linear combinations of entropies) generalizes to conditional entropies.

3.4.1 Examples

- $H(X|Y, Z) \leq H(X|Z)$
- $H(X, Y|Z) = H(Y|Z) + H(X|Y, Z)$
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$$H(X_1, \dots, X_n|Z) = H(X_1|Z) + H(X_2|X_1, Z) + H(X_3|X_2, X_1, Z) + \dots + H(X_n|X_1, \dots, X_{n-1}, Z)$$

- $H(X, Y|Z) \leq H(X|Z) + H(Y|Z)$
- $H(X_1, \dots, X_n|Z) \leq \sum_i H(X_i|Z)$

3.4.2 Proof sketch

To prove

- $H(X|Y, Z) \leq H(X|Z)$
- $H(X, Y|Z) = H(Y|Z) + H(X|Y, Z)$

It follows by

$$H(\cdot|Z) = \sum_c P(Z = c) \cdot H(\cdot|Z = c)$$

where $\cdot$ stands for any of the above entropies of conditional entropies. So any equality or inequality that holds conditioned on each event $Z = c$ separately also holds when we take the expectation over all possibilities for $Z = c$.

But we need to be careful when there are premises under which the equalities and inequalities are true.

3.4.3 A Word of Caution: Equalities Are Fragile

The averaging argument shows that any *universal* law of entropy — one that holds for every joint distribution, with no extra assumptions, such as subadditivity $H(A, B) \leq H(A) + H(B)$ or monotonicity $H(A) \leq H(A, B)$ — automatically generalizes to its conditional version. This is because such a law applies slice-by-slice to each conditional distribution $P(\cdot \mid Z = c)$, and inequalities (as well as equalities) are preserved under taking a nonnegative-weighted average over c .

However, an *equality* that we observe for a particular distribution is often not itself a universal law, but rather the *boundary case* of a universal inequality, attained only under some structural premise on the distribution (e.g. independence). Such equalities do *not* automatically survive conditioning, because the premise that produced them may fail to hold conditionally, even though it held unconditionally.

Example. Let $X, Y \sim \text{Bernoulli}(1/2)$ be i.i.d., and let $Z = X \oplus Y$. Then $X \perp Y$, so subadditivity is tight:

$$H(X, Y) = 2 = H(X) + H(Y).$$

But conditioning on Z destroys this independence: knowing Z and Y determines X exactly, and likewise for X, Y swapped, so

$$H(X, Y \mid Z) = 1, \quad H(X \mid Z) + H(Y \mid Z) = 1 + 1 = 2.$$

Subadditivity is still respected ($1 \leq 2$), but the equality has become a *strict* inequality. The premise $X \perp Y$ held unconditionally but fails conditionally on Z (in fact, X and Y become fully dependent given Z).

Heuristic. Whenever an identity you’ve derived is really the equality case of some universal inequality, ask *what property of the distribution made it tight* (independence, a Markov relation, a deterministic functional relation, etc.). That property — not the equality itself — is what must be checked *conditionally on* $Z = c$ for every c before you can assume the conditional equality still holds. The inequality is robust because it follows from the definitions alone; the equality is fragile because it depends on a fact about the specific distribution, and that fact need not survive conditioning.

4 Mutual Information

4.1 Setup

How did Sir Walter Raleigh measure the weight of the smoke of a cigar?

He weighted the cigar before he smoked it and subtracted the weight of the ashes that remained after he smoked the cigar. The difference was the weight of the smoke.

Scientifically, this is completely wrong. However, it's a nice way to think about mutual information.

4.2 Definition

$I(X; Y)$ = the information that X gives on Y

$$I(X; Y) = H(Y) - H(Y|X) \geq 0$$

Remember, this is a definition! And as a reminder, we know $I(X; Y) \geq 0$ because it follows from the fundamental inequality of conditional entropy.

The intuitive way to think about mutual information is as the reduction of uncertainty of Y due to knowledge about X .

4.3 Mutual Information is Symmetric

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(Y) - (H(X, Y) - H(X)) \\ &= H(Y) + H(X) - H(X, Y) \\ &= I(Y; X) = H(X) - H(X|Y) \end{aligned}$$

The information that Y gives on X equals to the information that X gives on Y . It's a bit counterintuitive, but the above holds and is very useful.

4.4 Some Simple Facts

- $I(X; Y) = 0$ iff X, Y are independent
- $I(X; X) = H(X)$
- $I(X; f(X)) = H(f(X)) \leq H(X)$ for any (deterministic) function f
- $I(X; Y, Z) \geq I(X; Y), I(X; Z)$

4.5 Conditional Mutual Information

$$\begin{aligned} I(X; Y|Z) &= H(Y|Z) - H(Y|X, Z) \\ &= H(Y, Z) - H(Z) - (H(Y, X, Z) - H(X, Z)) \\ &= H(Y, Z) + H(X, Z) - H(Z) - H(Y, X, Z) \\ &= I(Y; X|Z) \geq 0 \end{aligned}$$

As before, all equalities and inequalities (that involve linear terms) generalize to the case when conditioning on an additional random variable Z , as any such equality/inequality is an equality/inequality about entropies.

4.6 Chain Rule for Mutual Information

$$I(X_1, \dots, X_n; Y) = I(X_1; Y) + I(X_2; Y|X_1) + I(X_3; Y|X_1, X_2) + \dots + I(X_n; Y|X_1, X_2, \dots, X_{n-1})$$

and similarly, we can further condition everything on Z .

5 Concavity of the Entropy Function

Let $P_1 = (P_{1,1}, \dots, P_{1,m})$ and $P_2 = (P_{2,1}, \dots, P_{2,m})$ be two distribution on m elements. Let $0 \leq \lambda \leq 1$. Then,

$$\lambda \cdot H(P_1) + (1 - \lambda) \cdot H(P_2) \leq H(\lambda \cdot P_1 + (1 - \lambda) \cdot P_2)$$

5.1 Proof

Consider the following joint distribution of X, Y

	X		
Y	$\lambda \cdot p_{1,1}$	$\dots$	$\lambda \cdot p_{1,m}$
	$(1 - \lambda) \cdot p_{2,1}$	$\dots$	$(1 - \lambda) \cdot p_{2,m}$

$Y \in \{1, 2\}$ is such that $P(Y = 1) = \lambda$ and $P(Y = 2) = 1 - \lambda$. $X \in \{1, \dots, m\}$ is such that $P_{X|Y=1} = P_1$ and $P_{X|Y=2} = P_2$. Then we have

$$\begin{aligned} \lambda \cdot H(P_1) + (1 - \lambda) \cdot H(P_2) &= H(X|Y) \\ H(\lambda \cdot P_1 + (1 - \lambda) \cdot P_2) &= H(X) \end{aligned}$$

It's essentially just a restatement of the fundamental inequality of conditional entropy $H(X|Y) \leq H(X)$.

5.1.1 More explicitly

For $\lambda \cdot H(P_1) + (1 - \lambda) \cdot H(P_2) = H(X|Y)$, the logic is to perform some substitutions and then get the conditional entropy

$$\begin{aligned} &\lambda \cdot H(P_1) + (1 - \lambda) \cdot H(P_2) \\ &= P(Y = 1) \cdot H(P_1) + P(Y = 2) \cdot H(P_2) \\ &= P(Y = 1) \cdot H(X|Y = 1) + P(Y = 2) \cdot H(X|Y = 2) \\ &= \sum_{i \in \{1, 2\}} P(Y = i) \cdot H(X|Y = i) = H(X|Y) \end{aligned}$$

For $H(\lambda \cdot P_1 + (1 - \lambda) \cdot P_2) = H(X)$, the logic is substitute, then apply the chain rule, and then marginalize

$$\begin{aligned}
& H(\lambda \cdot P_1 + (1 - \lambda) \cdot P_2) \\
& H(P(Y = 1) \cdot P_1 + P(Y = 2) \cdot P_2) \\
& H(P(Y = 1) \cdot P_{X|Y=1} + P(Y = 2) \cdot P_{X|Y=2}) \\
& H(P(X, Y = 1) + P(X, Y = 2)) \\
& H(P(X)) = H(X) \quad \text{Just a notational change}
\end{aligned}$$

6 Markov Chains

We say that $X \rightarrow Y \rightarrow Z$ is a Markov Chain if $P(Z|X, Y) = P(Z|Y)$, that is, for every a, b, c (such that $P(X = a, Y = b) > 0$)

$$P(Z = c|X = a, Y = b) = P(Z = c|Y = b)$$

Intuitively, Y is generated from X and independent random bits and then Z is generated from Y and independent random bits (without knowing X). Each arrow can be viewed as a communication channel or some data processing. Y is a data processing of X and Z is a data processing of Y .

By the chain rule for probabilities, we have,

$$\begin{aligned}
P(X = a, Y = b, Z = c) &= \\
P(Y = b) \cdot P(X = a|Y = b) \cdot P(Z = c|X = a, Y = b)
\end{aligned}$$

Thus, $P(Z = c|X = a, Y = b) = P(Z = c|Y = b)$ is equivalent to

$$P(X = a, Y = b, Z = c) = P(Y = b) \cdot P(X = a|Y = b) \cdot P(Z = c|Y = b)$$

for every a, b, c . This is an equality, so if we have a Markov chain we get $P(Z = c|X = a, Y = b) = P(Z = c|Y = b)$, and if we have $P(Z = c|X = a, Y = b) = P(Z = c|Y = b)$, we have a Markov chain.

Thus,

1. $Z \rightarrow Y \rightarrow X$ is also a Markov Chain
2. Conditioned on Y , we have that X, Z are independent
3. $I(X; Z|Y) = 0$

6.0.1 Deriving points 1 and 2

Points 1 and 2 are quite deep and not always immediate. Here is some justification.

Point 2: Conditional independence of X, Z given Y . Dividing the factorization by $P(Y = b)$ gives, for every a, b, c ,

$$\begin{aligned} P(X = a, Z = c \mid Y = b) &= \frac{P(X = a, Y = b, Z = c)}{P(Y = b)} \\ &= \frac{P(Y = b) \cdot P(X = a \mid Y = b) \cdot P(Z = c \mid Y = b)}{P(Y = b)} \\ &= P(X = a \mid Y = b) \cdot P(Z = c \mid Y = b). \end{aligned}$$

That is, $P(X, Z \mid Y) = P(X \mid Y) \cdot P(Z \mid Y)$, so conditioned on Y , X and Z are independent.

Point 1: The reverse chain $Z \rightarrow Y \rightarrow X$. By the chain rule for probabilities, $P(Y = b, Z = c) = P(Y = b) \cdot P(Z = c \mid Y = b)$ for every b, c . Dividing the original factorization by this gives, for every a, b, c ,

$$\begin{aligned} P(X = a \mid Y = b, Z = c) &= \frac{P(X = a, Y = b, Z = c)}{P(Y = b, Z = c)} \\ &= \frac{P(Y = b) \cdot P(X = a \mid Y = b) \cdot P(Z = c \mid Y = b)}{P(Y = b) \cdot P(Z = c \mid Y = b)} \\ &= P(X = a \mid Y = b). \end{aligned}$$

That is, $P(X \mid Y, Z) = P(X \mid Y)$, which is exactly the defining condition for $Z \rightarrow Y \rightarrow X$ to be a Markov chain.

6.1 Data Processing Inequality

If $X \rightarrow Y \rightarrow Z$ is a Markov Chain, then

$$I(X; Z) \leq I(X; Y)$$

or equivalently

$$H(X|Z) \geq H(X|Y)$$

The equivalence is because

- $I(X; Z) = H(X) - H(X|Z)$
- $I(X; Y) = H(X) - H(X|Y)$

The theorem is very intuitive in the data processing interpretation: if we further process the data, we reduce the information that it gives about the source.

6.1.1 Proof

If $X \rightarrow Y \rightarrow Z$ is a Markov Chain, then $Z \rightarrow Y \rightarrow X$ is also a Markov Chain. We can write

$$\begin{aligned} P(X|Y, Z) &= P(X|Y) \\ \mathbb{E}[\log P(X|Y, Z)] &= \mathbb{E}[\log P(X|Y)] \\ H(X|Y, Z) &= H(X|Y) \end{aligned}$$

Hence,

$$H(X|Z) \geq H(X|Y, Z) = H(X|Y)$$

7 Relative Entropy

Suppose $p = (p_1, \dots, p_m)$ and $q = (q_1, \dots, q_m)$ are two distributions on m elements. The relative entropy¹ is

$$D(p||q) = \sum_i p_i \log \left(\frac{p_i}{q_i} \right) \geq 0$$

We read it as “D” of “p” relative to “q”.

Relative entropy may be infinite. We can prove the inequality by a convexity argument. When q is uniform, it is the same as Shannon Entropy (up to a sign and an additive constant). We will learn later in the course what $D(p||q)$ really is.

8 Some exotic entropy functions

8.1 Entropy of a Random Variable over the Integers

Let X be a discrete random variables (possibly infinite number of states), say over the integers

$$H(X) = - \sum_{i=-\infty}^{\infty} P(X=i) \log P(X=i) = - \sum_i P_i \log P_i$$

as in the finite case. The sum may be infinite, but otherwise all equalities and inequalities generalize.

8.2 Entropy of a Continuous Random Variable

If X is a continuous random variable, we define

$$H(X) = - \int f(x) \log f(x) dx$$

where $f(x)$ is the probability density function (in analogy to the finite case). We assume that X has a “nice” probability density function and that the integral is well defined. This is also called Differential Entropy. It may be negative. Equalities generalize, *but inequalities not necessarily*.

¹We will prove that this is always equal to or greater than 0 via Jensen inequality.

8.2.1 Differential Entropy may be Negative

Note that $H(X)$ may be infinite, and also negative. For example $f(x) = \frac{1}{a}$ on the interval $[0, a]$. Then

$$-\int_0^a f(x) \log f(x) dx = \log a$$

which is negative when $a < 1$. This destroys all inequalities (but equalities remain the same).

8.3 Relative Entropy in the Continuous Case

If f, g are sufficiently nice probability density functions

$$D(f\|g) = \int f(x) \log \left(\frac{f(x)}{g(x)} \right) dx > 0$$

9 Approximating Binomial Coefficients (informal)

Let $k \leq n$ be integers. Let $p = \frac{k}{n}$. We have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

Stirling's approximation is $\ln(m!) \approx m \ln m - m$. Thus,

$$\ln \binom{n}{k} \approx (n \ln n - n) - (k \ln k - k) - ((n-k) \ln(n-k) - (n-k))$$

which then can be written as

$$\begin{aligned} \ln \binom{n}{k} &\approx n \ln n - k \ln k - (n-k) \ln(n-k) = -k \ln \left(\frac{k}{n} \right) - (n-k) \ln \left(\frac{n-k}{n} \right) \\ &= -np \ln p - n(1-p) \ln(1-p) = n \cdot (-p \ln p - (1-p) \ln(1-p)) \end{aligned}$$

Hence, we have

$$\log \binom{n}{k} \approx n \cdot (-p \ln p - (1-p) \ln(1-p)) = n \cdot H(p, 1-p)$$

and

$$\binom{n}{k} \approx 2^{n \cdot H(p, 1-p)}$$

9.1 Approximating Multinomial Coefficients

Let $(p_1, \dots, p_m)$ be a distribution and thus $\sum_{i=1}^m p_i = 1$. Let $n \in \mathbb{N}$ and let $(n_1, \dots, n_m)$ be such that $n_i = np_i$ and thus $\sum_{i=1}^m n_i = n$. Consider the multinomial coefficient

$$\frac{n!}{n_1! \cdot n_2! \cdots n_m!} = \frac{n!}{(np_1)! \cdot (np_2)! \cdots (np_m)!}$$

Given the work that we did in the last subsection, it follows that

$$\frac{1}{n} \cdot \log \left(\frac{n!}{n_1! \cdot n_2! \cdots n_m!} \right) \rightarrow - \sum_{i=1}^m p_i \log p_i = H(p_1, \dots, p_m)$$

where it follows that

$$\binom{n}{n_1, \dots, n_m} \approx 2^{n \cdot H(p_1, \dots, p_m)}$$